

FMUSER FUQ-400B Modular Headend Processor 6-Slot Configurable ASI-IP I-O MPEG-2-H.264-H.265 DVB-ISDB Encoding Decoding Transcoding Multiplexing Modulating For CATV Systems



Product overview

The FMUSER FUQ-400B is a modular head-end processor in a 1U chassis, supporting encoding, decoding, transcoding, multiplexing, descrambling, and modulating via six configurable slots. It handles signals through IP input/output via GE1 and GE2 ports and ASI output. Controlled via a web-based interface, it offers optional redundancy power supply. Ideal for CATV systems requiring flexible and cost-effective signal processing in head-end applications.

Key Features

Supports flexible module configuration with up to six slots for encoding, decoding, transcoding, multiplexing, descrambling, and modulating,

Features IP input/output through GE1 and GE2 ports and ASI output,

Offers redundancy power supply with hot or non-hot plugging options,

Enables web-based management for remote configuration and updates

Specifications

4 CVBS/SDI Encoding Module(DX214B)

**OSD Insertion**

Module Specifications:

Input: 4 CVBS (DB9 to RCA) or 4 SDI (BNC)

Output: 1MPTS and 4 SPTS output over UDP/RTP, unicast and multicast

Video Encoding:

Video format: MPEG-2, MPEG4 AVC/H.264

Image format: PAL, NTSC SD signal (Only for CVBS input)

Resolution:

*Input: 720*576 @50i*

*Output: 720*576/352*288/320*240/320*180/176*144/160*120/160*90@50Hz*

*Input: 720*480 @60i*

*Output: 720*480/352*288/320*240/320*180/176*144/160*120/160*90@60Hz*

Rate Control: CBR/VBR

GOP structure: IPPP, IBPBP, IBBPB, IBBBBP

Video bitrate: 0.5~5Mbps

Audio Encoding:

Audio format: MPEG1 Audio Layer 2, LC-AAC, HE-AAC

Sampling rate: 48KHz

Bits per sample: 32-bit

Bit-rate: 48-384Kbps each channel

*Support **Logo, Caption, QR Code insertion***

4 HDMI/SDI Encoding Module (DX224V)





Module Specifications:

Input: 4×SDI/HDMI (1.4) input, HDCP 1.4

Output: 1 MPTS and maximum 4 SPTS output over UDP/RTP/RTSP

Video Encoding:

Video format: **HEVC/H.265** & MPEG 4 AVC/H.264

Resolution:

HDMI:

3840×2160_30P, 3840×2160_29.97P;

(Encoding 2 CHs per module for H.265, and encoding 1 CH for H.264)

1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P;

(Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264)

1280×720_60P, 1280×720_59.94P, 1280×720_50P

(Encoding 4 CHs per module for H.264 and H.265)

SDI:

1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P;

(Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264)

1280×720_60P, 1280×720_59.94P, 1280×720_50P

(Encoding 4 CHs per module for H.264 and H.265)

Input: 1920×1080_60i, 1920×1080_59.94i, 1920×1080_50i

Output: 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P

(Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264)

Chroma: 4:2:0 Bit rate: 0.5Mbps~20Mbps (each channel)

Rate Control: CBR/VBR GOP structure: IBBP, IPPP

Audio Encoding:

Audio format: MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough

Sampling rate: 48KHz

Bit-rate (each channel):

48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC)

24 Kbps~128 Kbps (HE-AAC)

18 Kbps~56 Kbps (HE-AAC V2)

Audio Gain: 0~255

4 HDMI Encoding Module(DX224H/HV)



Module Specifications:

Input: 4×HDMI (1.4) input, HDCP 1.4

Output: 1 MPTS and 4 SPTS output over UDP/RTP/RTSP; IPv4, IPv6 output

Video Encoding:

Video format: HEVC/H.265 & MPEG 4 AVC/H.264---DX 224H

HEVC/H.265---DX224HV

Resolution:

1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P;

1280×720_60P, 1280×720_59.94P, 1280×720_50P

Input: 1920×1080_60i, 1920×1080_59.94i, 1920×1080_50i

Output: 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P

Chroma: 4:2:0 Bit rate (each channel): 0.5Mbps~20Mbps (H.265)

4 Mbps~20Mbps (H.264)

Rate Control: CBR/VBR GOP structure: IBBP, IPPP

Audio Encoding:

Audio format: MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough

Sampling rate: 48KHz

Bit-rate (each channel):

48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC)

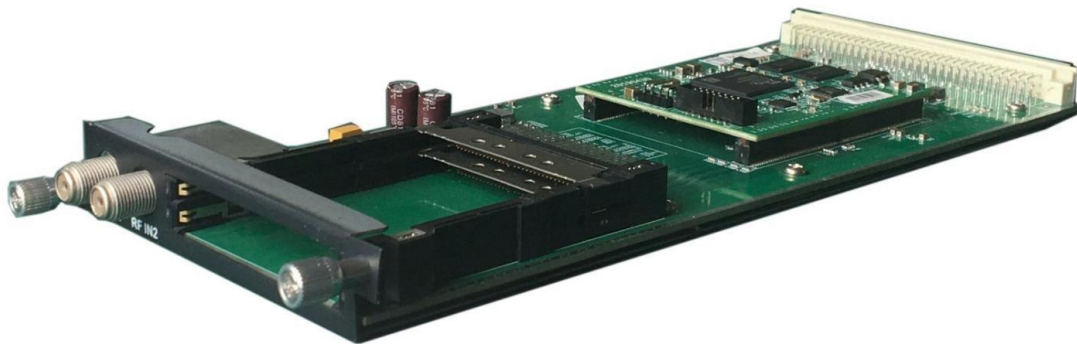
24 Kbps~128 Kbps (HE-AAC)

18 Kbps~56 Kbps (HE-AAC V2)

Audio Gain: 0~255

Support **Logo, QR Code insertion**--Optional as per order

2 Tuner Descrambling Module (DX902A)



Module Specifications:

Stream in: 2 Tuner input, F Type

Stream out: 1 MPTS output output over UDP, Unicast/Multicast, SFP/RJ45

DVB-CI: 2 independent common interface slots

Standard: DVB-S/S2/**S2X**

DVB-S Input Frequency: 950-2150MHz

Symbol Rate: QPSK: 0.5~45 Msps

FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8

DVB-S2 Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK/16APSK: 0.5~45 Msps; 32APSK: 0.5~40 Msps

FEC Demodulation: QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10

32APSK: 3/4, 4/5, 5/6, 8/9

DVB-S2X Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK/16APSK: 0.5~45 Msps; 8APSK/32APSK: 0.5~40 Msps

FEC Demodulation: QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18

8APSK: 5/9-L, 26/45-L

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90

32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9

Signal Strength: -65~ -25dBm

Support Diseqc function (For DX902A)

Multiplexing:

Maximum PID Remapping: 256 output pids

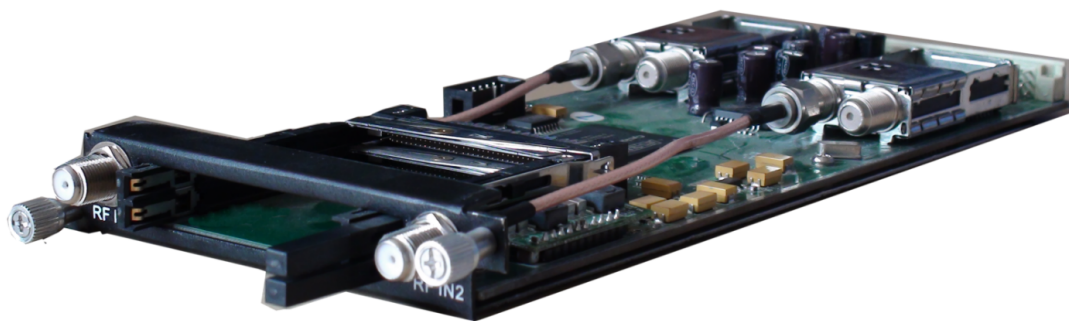
Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically

Descrambling:

CAM/CI Quantity: 2

BISS Mode: Mode 1, Mode E; up to 120Mbps (Optional as required)

2 Tuner Descrambling Module (DX912)



Module Specifications:

Stream in: 2 Tuner input, F Type

Stream out: 1 MPTS output over UDP, Unicast/Multicast, SFP/RJ45

DVB-CI: 2 Independent common interface slots

Standard: DVB-C (J.83 A/C)/J.83B/ DVB-T/(T)/ISDB-T switchable

Standard: DVB-C (J.83 A/C); J.83B

Input Frequency: 60MHz~890MHz

Symbol rate: 1000~9000Ksps

Constellation: 16/32/64/128/256 QAM; 64/256 QAM for J.83B

Standard: DVB-T/(T)

Frequency In: 60MHz~890MHz

Bandwidth: 5/6/7/8M bandwidth

PLP Index: 0~255 (for DVB-T))

Standard: ISDB-T

Input Frequency: 60-890MHz

Signal Strength: -65~ -25dBm

Multiplexing:

Maximum PID Remapping: 256 output pids

Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically

Descrambling:

CAM/CI Quantity: 2

BISS Mode: Mode 1, Mode E; up to 120Mbps (Optional as required)

4 HDMI Decoding Module (DX714)



Module Specifications:

Input: 4 IP (MPTS/SPTS) input from data port over UDP&RTP, Unicast/Multicast

Output: 4 HDMI output

Decoding:

Video/Audio Out: 4 HDMI output with 1 channel stereo audio embedded in each port

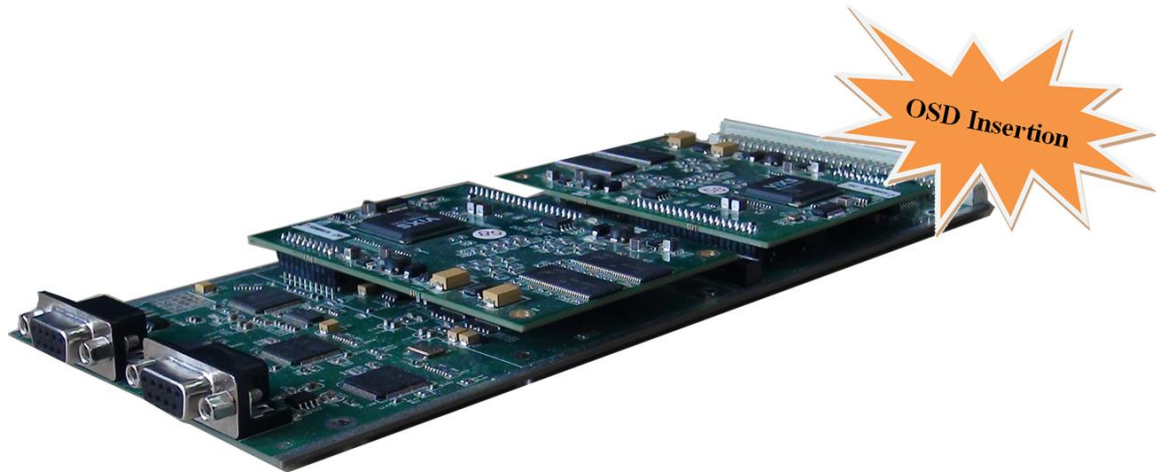
Video Format: MPEG-2, MPEG-4 AVC/H.264, HEVC/H.265, AVS, AVS+

Audio Format: MPEG 1 Layer 2, LC-AAC, HE-AAC, AC3 (2.0)

Resolution: 480i, 480p, 576i, 576p, 1280×720_50p, 1280×720_60p, 1920×1080_50i, 1920×1080_50p, 1920×1080_60i, 1920×1080_60p

Support manually upscale/downscale resolution

4 CVBS Encoding Module(DX214/DX214A)



Module Specifications:

Input: 4 CVBS video, 4 Stereo Audio (DB9 to RCA)

Output: 1MPTS and 4 SPTS output over UDP/RTP, unicast and multicast, SFP/RJ45

Video Encoding:

Video format: MPEG-2 (4:2:0)

Image format: PAL, NTSC SD signal

*Input resolution: 720×480_60i, 544×480_60i, 352×480_60i, 352×240_60i,
320×240_60i, 176×240_60i, 176×120_60i, 720×576_50i,
704×576_50i, 640×576_50i, 352×288_50i, 320×288_50i,
176×288_50i, 176×144_50i*

GOP structure: IP, IBP, IBBP, IBBBP

Video bit rate: 0.5Mbps~8Mbps per channel

Support CC (closed caption)

Audio Encoding:

Audio format: MPEG-1 Layer 2, DD AC3 (2.0)

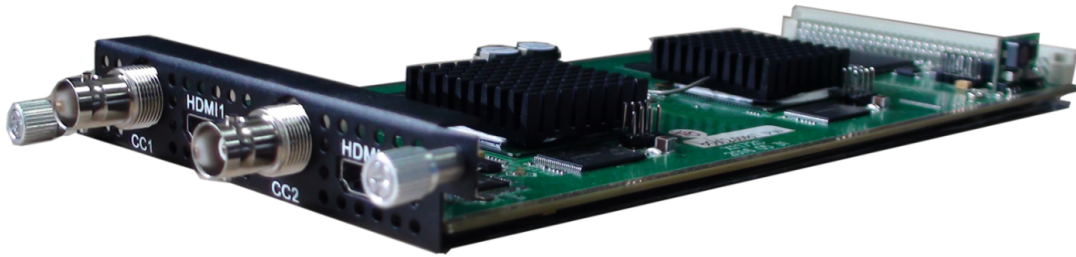
Sampling rate: 48KHz

Resolution: 24-bit

Audio bitrate: 128/192/256/320/384kbps each channel

Support Logo, Caption, QR Code insertion (for DX214A only)

2 HDMI Encoding/Transcoding Module (DX202A)



Module Specifications:

*Input: 2*HDMI, 2*BNC for CC (Closed Caption) input*

*Output: 1*MPTS output over UDP, Unicast/Multicast*

Video Encoding:

Video format: MPEG2 & MPEG4 AVC/H.264

Input resolution:

*1920*1080_60P, 1920*1080_50P, 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i*

Rate control mode: CBR/VBR

Aspect ratio: 16:9, 4:3

Video bitrate: 0.8~19Mbps for H.264 encoding

1~19.5Mbps for MPEG-2 encoding

Support CC (closed caption)

Audio Encoding:

Audio format: MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC,

Dolby Digital AC3 (2.0) encoding (Optional); AC3 (2.0/5.1) passthrough

Sampling rate: 48KHz

Audio bitrate: 64Kbps-320kbps each channel

Video Tanscoding:

*2*MPEG2 HD → 2*MPEG2/H.264 HD; 2*MPEG2 HD → 2*MPEG2/H.264 SD;*

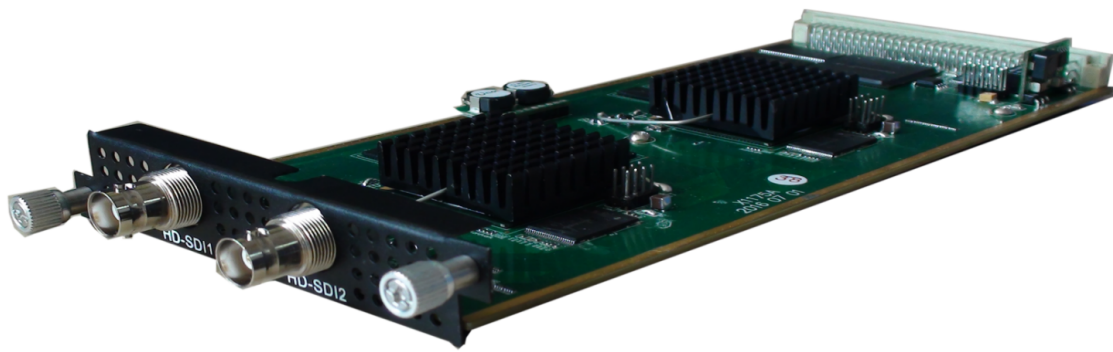
2 H.264 HD → 2*MPEG2/H.264 HD; 2* H.264 HD → 2*MPEG2/H.264 SD;*

*4 *MPEG2 SD → 4 *MPEG2/H.264 SD; 4* H.264 SD → 4 *MPEG2/H.264 SD*

Audio Tanscoding:

MPEG-1 Layer 2, AC3 (Optional) and AAC any-to-any

2 SDI Encoding/Transcoding Module (DX202A-D)



Module Specifications:

*Input: 2*HD-SDI*

*Output: 1*MPTS output over UDP, Unicast/Multicast*

Video Encoding:

Video format: MPEG2 & MPEG4 AVC/H.264

Input resolution:

*1920*1080_60P, 1920*1080_50P, 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i*

Rate control mode: CBR/VBR

Aspect ratio: 16:9, 4:3

Video bitrate: 0.8~19Mbps for H.264 encoding;

1~19.5Mbps for MPEG-2 encoding

Support CC (closed caption)

Audio Encoding:

Audio format:

MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC,

Dolby Digital AC3 (2.0) encoding (Optional), AC3 (2.0/5.1) passthrough

Sampling rate: 48KHz

Audio bitrate: 64Kbps-320kbps each channel

Video Tanscoding:

*2*MPEG2 HD → 2*MPEG2/H.264 HD; 2*MPEG2 HD → 2*MPEG2/H.264 SD;*

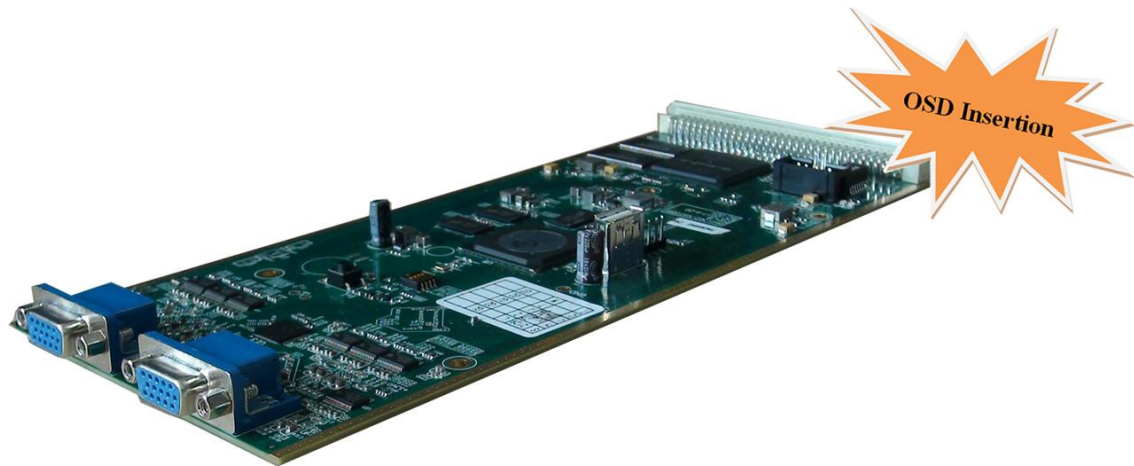
2 H.264 HD → 2*MPEG2/H.264 HD; 2* H.264 HD → 2*MPEG2/H.264 SD;*

*4 *MPEG2 SD → 4 *MPEG2/H.264 SD; 4* H.264 SD → 4 *MPEG2/H.264 SD*

Audio Tanscoding:

MPEG-1 Layer 2, AC3 (Optional) and AAC any-to-any

8 CVBS Encoding Module(DX218S)



Module Specifications:

Input: 8 CVBS video, 8 Stereo Audio (DB15 to RCA)

Output: 1 MPTS and 8 SPTS output over UDP/RTP, unicast and multicast,

Video Encoding:

Video format: MPEG4 AVC/H.264

Image format: PAL, NTSC SD signal

Resolution: 720×576i, 720×480i

Rate Control: CBR/VBR

GOP structure: IPP

Video bitrate: 1~7Mbps each channel

Audio Encoding:

Audio format: MPEG-1 Layer 2

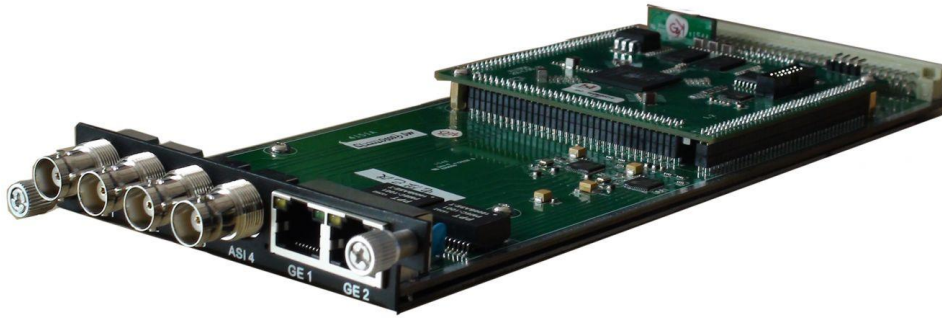
Sampling rate: 48KHz

Resolution: 24-bit

Bit-rate: 64/128/192/224/256/320/384Kbps each channel

Support Logo, Caption, QR Code insertion (Language Supported: ■■, English, ■■■■, for more languages please consult us...)

4 ASI/IP Multiplexing Module (DX504)



Module Specifications:

ASI inputs/outputs: 4 ASI bi-direction, BNC 75Ω

IP inputs/outputs: 2 Ethernet Port(100/1000M)

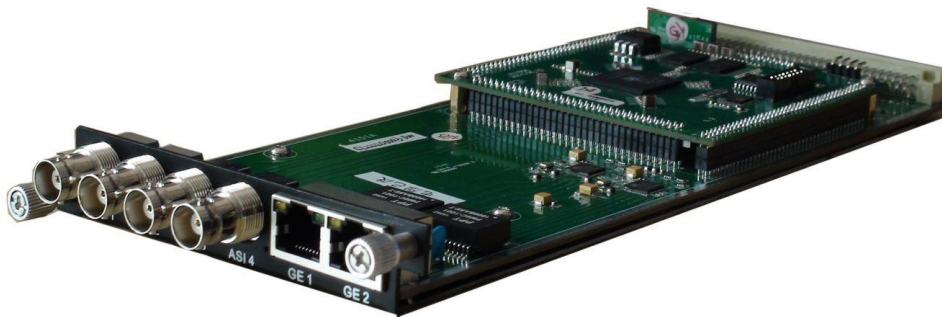
Stream connector input

Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically

Stream In: maximum 4 ASI input, 256×2 IP input

Stream Out: maximum 4 ASI output, 4 IP output

4/8 CH EAS Multiplexing Module (DX504E/DX508E)



Module Specifications:

ASI input: 1 ASI input (SPTS), BNC 75Ω

IP input: 256 IP input thru 1 GE1 (100/1000M)

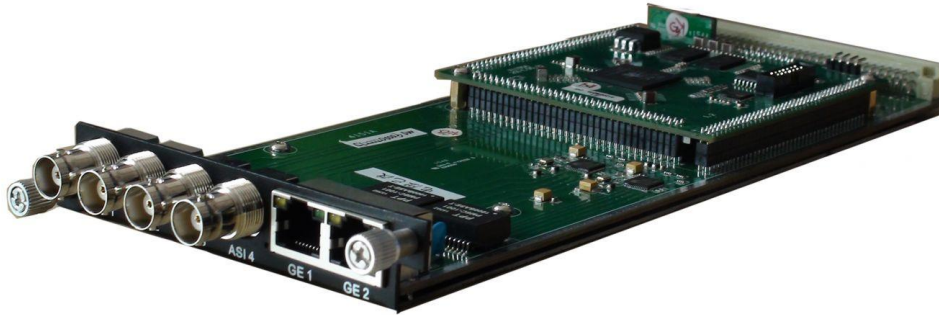
EAS Source: ASI or IP (the 256th IP) (ASI&IP should be SPTS, both can't mux, source Bitrate≤10Mbps)

Re-multiplexing: PID remapping, PCR correction (only for IP), generate PSI/ SI table automatically

Stream Out: 4 IP output thru GE1, maximum 16 programs each channel--DX504E

Stream Out: 8 IP output thru GE1, maximum 8 programs each channel--DX508E

IP Adapter Module (DX514)



Module Specifications:

IP Ports: GE1/GE2 (100/1000M self-adaption)

IP Input: 10 IP input through GE1/GE2

IP Output: 10 IP output switch to other module

ASI Ports: 4 ASI bi-direction, BNC 75Ω

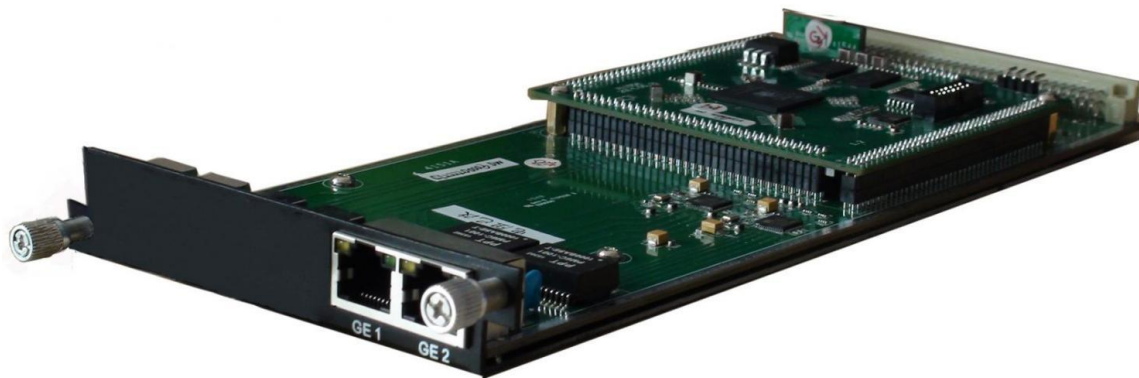
Select ASI port as input, IP output thru GE1/GE2 as mirror of ASI input

Select ASI port as output, ASI (1/2/3/4) output as mirror of IP in (IP 1/2/3/4)

Support PCR correction, **IP De-jitter**

Suggest match with DX902/DX912 module to descramble IP

IP Multiplexing Module (DX506)



Module Specifications:

IP input: 512 SPTS or MPTS input over UDP, RTP, Unicast and Multicast thru GE1 Ethernet Port (100/1000M)

Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically

IP Out: 512 SPTS output over UDP, RTP, Unicast and Multicast through GE2 Ethernet Port (100/1000M)

16/32 QAM Modulating Module (DX316/DX332)



Module Specifications:

Data input: Max 512 IP input over UDP/RTP, 2 GE Ports (RJ45/SFP) (DX316)

Max 1024 IP input over UDP/RTP, 2 GE Ports (RJ45/SFP) (DX332)

128 IP input through front panel GE2 port or internal IP output

Data output: 16 or 32 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP)

Trans Rate: Max 840Mbps/GE Port

RF output (F type): 16/32 channels of multiplexing, scrambling and modulation.

Multiplexing:

Maximum PID Remapping: 180 output pids per channel for DX316. 256 output pids per channel for DX332.

Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/SI table automatically

Scrambling:

Maximum simulcrypt CA: 4

Standard: ETR289, ETSI 101 197, ETSI 103 197

Connection: Local/remote connection

Modulation:

Standard: EN300 429/ITU-T J.83A/B (DVB-C)

MER: ≥ 40 dB

RF frequency: 50~960MHz, 1KHz step

RF output level: -20~+10dBm (87~117 dB μ V), 0.1dB step for all carriers

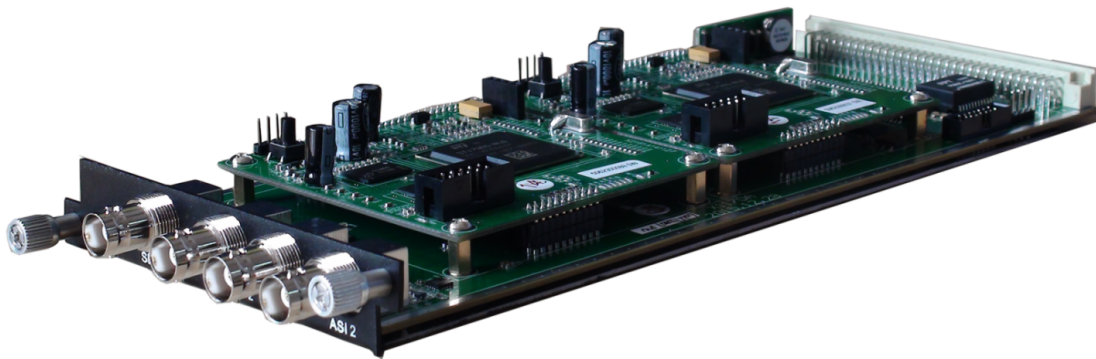
Symbol Rate: 5.0Msps~7.0Msps, 1ksps step

Constellation: 16/32/64/128/256QAM

DX316 Output: 16 non-adjacent carrier outputs within 192M bandwidth

DX332 Output: 32 non-adjacent carrier outputs within 384M bandwidth

2 HD-SDI Decoding Module(DX702)



Module Specifications:

ASI input/output: 2 ASI bi-direction, BNC 75Ω

IP input: 2 IP input from data port

IP output: 1 MPTS output over UDP, Unicast/Multicast

Decoding:

Video/Audio Out: 2 HD/SD SDI output

Video Format: MPEG-2, MPEG-4 AVC/H.264

Resolution: 480i, 480p, 576i, 576p, 720p@50/59.94/60, 1080i@50/59.94/60

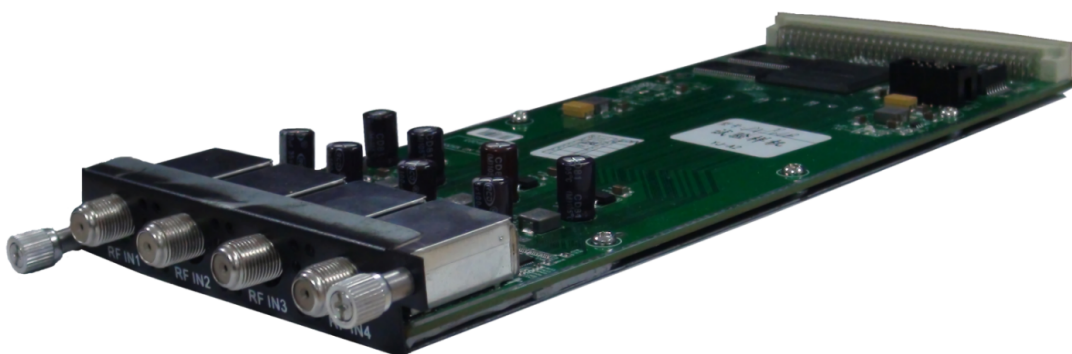
Chroma: 4:2:0

Audio Format: MPEG1 Layer2, LC-AAC, HE-AAC, AC3 (2.0/5.1), AC3 Passthrough,

Support Dual Audio Out

Support **CC/Subtitle**

4 FTA Tuner Module (DX904B)



Module Specifications:

Stream in: 4 Tuner input, F Type

Stream out: 5 MPTS out over UDP, unicast/multicast

Standard: DVB-S/S2/S2X

Tuner Section

DVB-S

Input Frequency: 950-2150MHz

Symbol Rate: QPSK: 0.5~45 Msps

FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8

DVB-S2

Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK/16APSK: 0.5~45 Msps;

32APSK: 0.5~40 Msps

FEC Demodulation: QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10

32APSK: 3/4, 4/5, 5/6, 8/9

DVB-S2X

Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK/16APSK: 0.5~45 Msps;

8APSK/32APSK: 0.5~40 Msps

FEC Demodulation:

QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18

8APSK: 5/9-L, 26/45-L

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90

32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9

Signal Strength: -65 ~ -25dBm

Support Diseqc function

Multiplexing:

Max number of managing PID: 256

Function: PID remapping (automatically/ manually), Accurate PCR adjust, PID pass-through

Equipment Specifications:

Base Unit Parameters

Dimension(W×L×H): 482mm×410mm×44mm
Environment: 0~45℃(work); -20~80℃(Storage)
Power requirements AC 110V± 10%, 50/60Hz, AC 220 ± 10%, 50/60Hz

Parameters Comparison:

	DHP400	DHP400A
IP Data Transport	One-way Transport: GE1/GE2 support output maximum 8MPTS & 512SPTS	Bi-directional Transport: GE1/GE2 support IP data input and output
TS Processing Number	Support 1 TS (MPTS/SPTS) input from each module, and maximum bitrate is 350Mbps	Support maximum 512 TS (MPTS/SPTS) input from each module, and maximum bitrate is 350Mbps
Multiplexing	Support multiplexing function: It can multiplex TSs from different modules to one TS and output through one module or GE1/GE2 port	Doesn't support multiplexing function: It can combine TSs from different modules and output these TSs through one module or GE1/GE2 port
Output Per Module	1 MPTS after multiplexing	1 or multiple MPTS/SPTS

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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